January 6, 2025

Technical Bulletin TB102

AQUACELL load capacity testing results by TRI Environmental

Objective

The objective was to evaluate the structural integrity and performance of AquaCell stormwater crates in two configurations (Standard & Extra Strong) under lateral compression.

Test Method

The testing was performed following a modified version of the ASTM D2412 standard, which is the Standard Test Method for Determination of External Loading Characteristics of Plastic Pipe by Parallel-Plate Loading. The modifications included:

- **Loading Rate:** Compressive load applied at a constant rate of 2.0% of the chamber height per minute.
- **Standards Referenced:**
  - ASTM F2418: Standard Specification for Polypropylene (PP) Corrugated Wall Stormwater Collection Chambers.
  - ASTM F2922: Standard Specification for Polyethylene (PE) Corrugated Wall Stormwater
  - Collection Chambers.

Test Results

1) **Standard Configuration:**
   a) Average maximum compressive stress applied laterally: 17.2 psi
   b) Tested with two side plates to be considered a complete unit with columns.

Product	Test Specimen	Maximum Compressive Load(lb)	Maximum Compressive Stress (psi)	Extension at Max Load (in)
Aquacell Standard Configuration - Lateral	1	14,239	18.0	0.922
	2	13,107	16.6	0.873
	3	13,590	17.2	0.994
	Avg.	**13,645**	**17.2**	**0.930**

2) **Extra Strong Configuration:**
   a) Average maximum compressive stress applied laterally: 26.4 psi
   b) Tested with two side plates to be considered a complete unit with columns.

Product	Test Specimen	Maximum Compressive Load(lb)	Maximum Compressive Stress (psi)	Extension at Max Load (in)
Aquacell Extra Strong Configuration - Lateral	1	22,453	26.4	1.166
	2	22,354	26.3	1.210
	3	22,551	26.5	1.264
	Avg.	**22,453**	**26.4**	**1.213**

Discussion

The Aquacell stormwater crates demonstrated robust performance under lateral compression. The Aquacell Standard (SD) configuration showed an average maximum compressive load of 13,645 lbs and an average maximum compressive stress of 17.2 psi. The Aquacell Extra Strong (EX) configuration exhibited an average maximum compressive load of 22,453 lbs and an average maximum compressive stress of 26.4 psi. These results indicate that both configurations possess the structural integrity required for effective stormwater management applications, with the excpectation that backfill, and finished pavement will further improve load distribution and performance.

Conclusion

Based on testing, and as supplement to the TB-101 Technical Bulletin regarding vertical compression strength, the AQUACELL system can be utilized in the following load classifications

a) AASHTO H-20, HS-20, H-25 (Standard Configuration)
b) AASHTO HS-25 (Extra Strong Configuration)

Note

1) AASHTO refers to the American Association of State Highway and Transportation Officials.
2) Testing completed at TRI Environmental Inc., Austin, Texas.
    a. Results referenced from TRI log number 24-005619 dated December 19, 2024.

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